HumanoidMimicGen

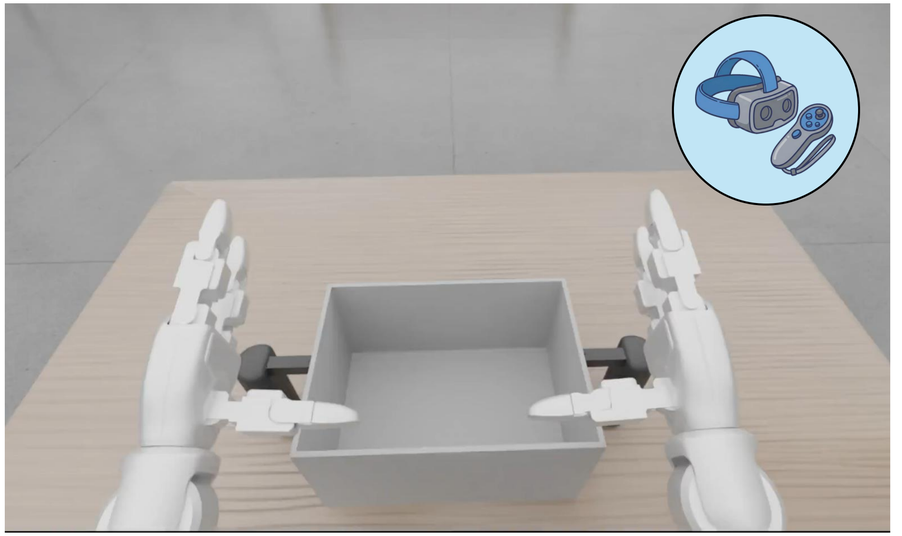
Data Generation for Loco-Manipulation via Whole-Body Planning and Adaptation

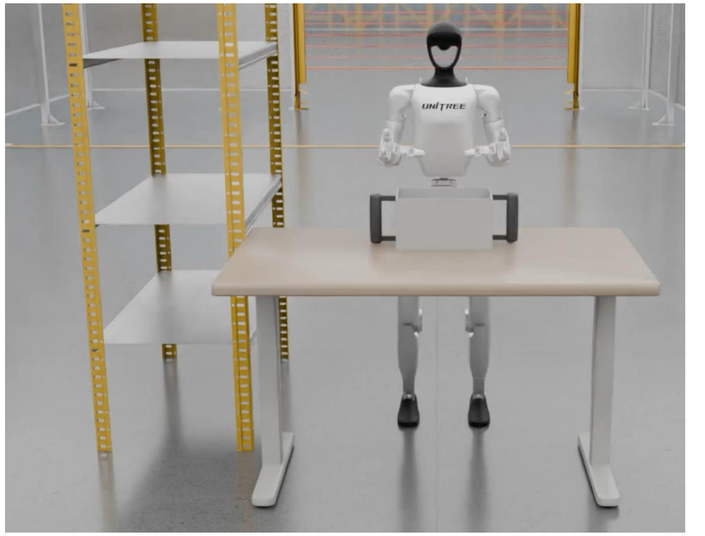
Paper-ID 851

HumanoidMimicGen automatically generates large humanoid imitation learning datasets from a few human demos

1x human demos

>1000 generated demos

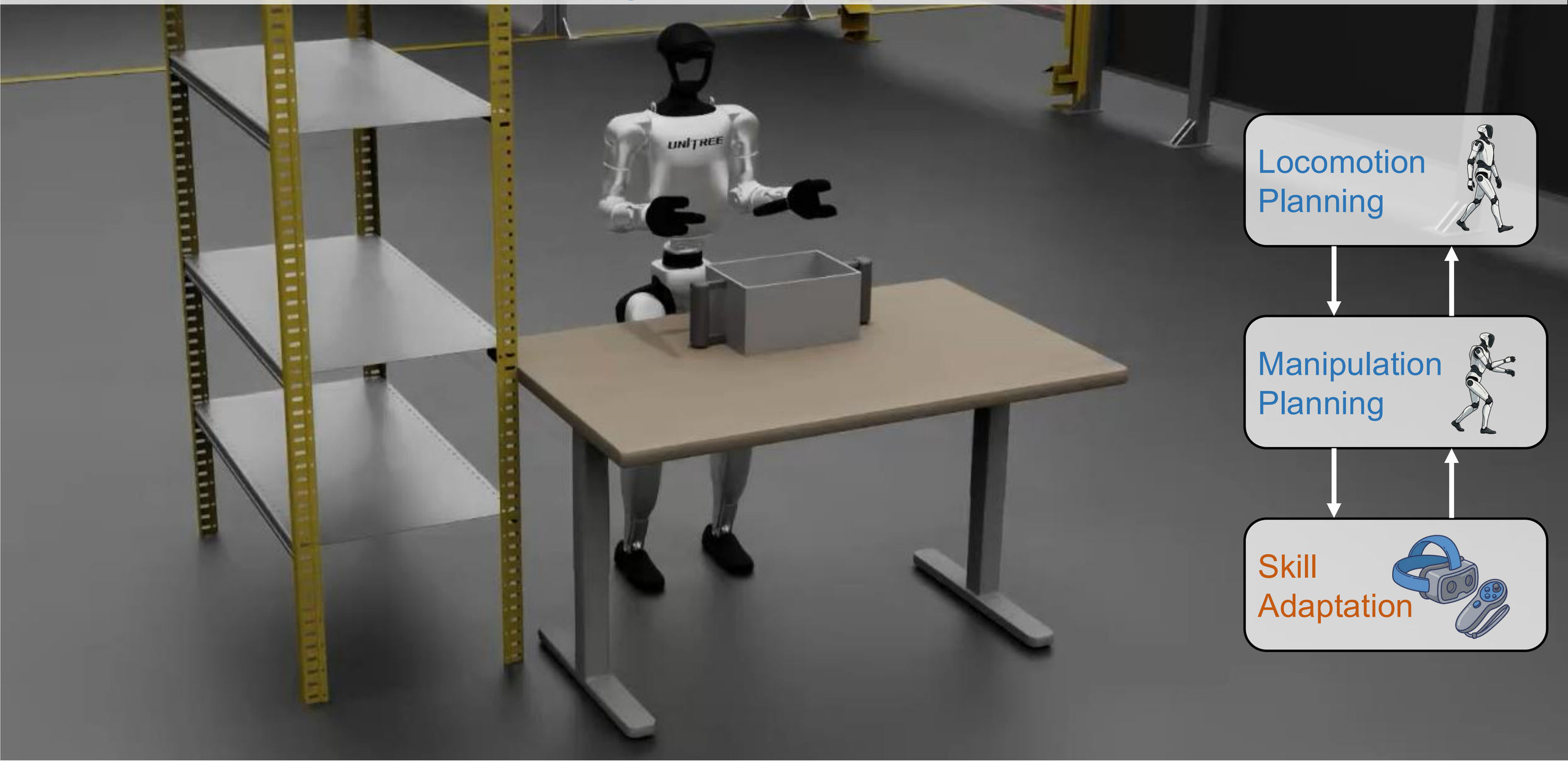




Human collects a single teleoperated demonstration

Humanoid-MG generates 1000x more with Whole-Body Planning and Adaptation

HumanoidMimicGen generates data by interleaving locomotion planning, manipulation planning, and demonstrated skill adaptation



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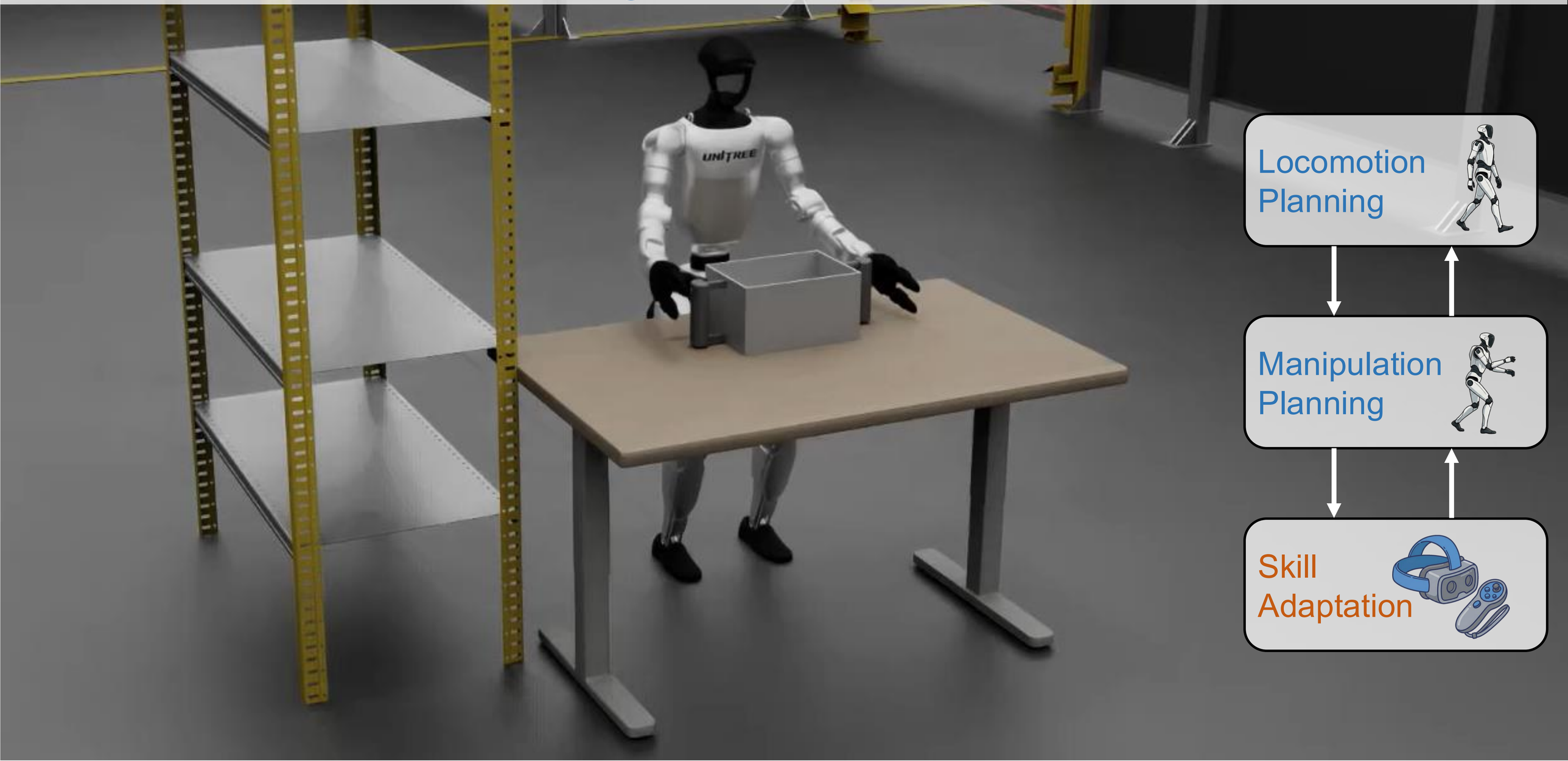

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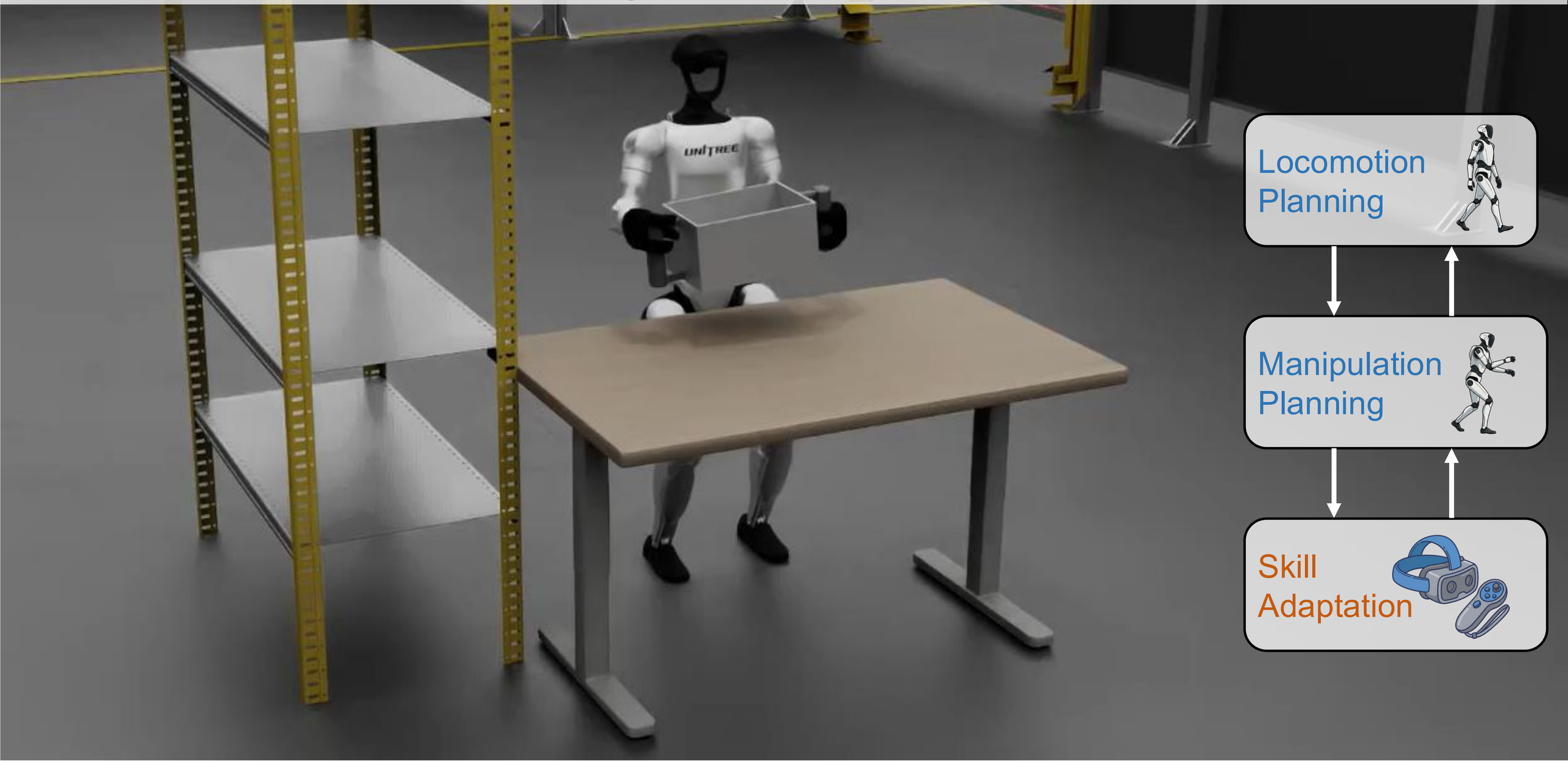

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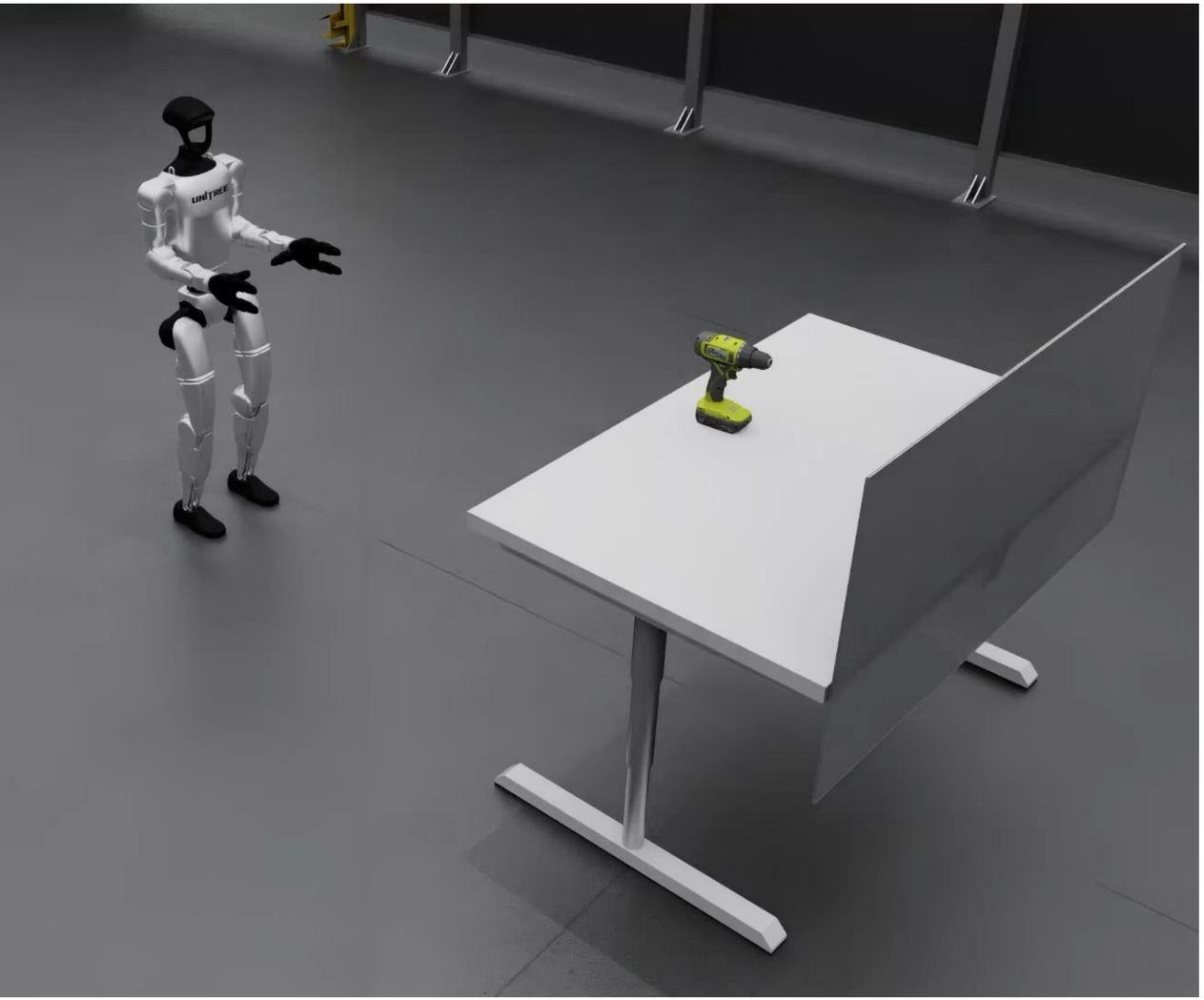
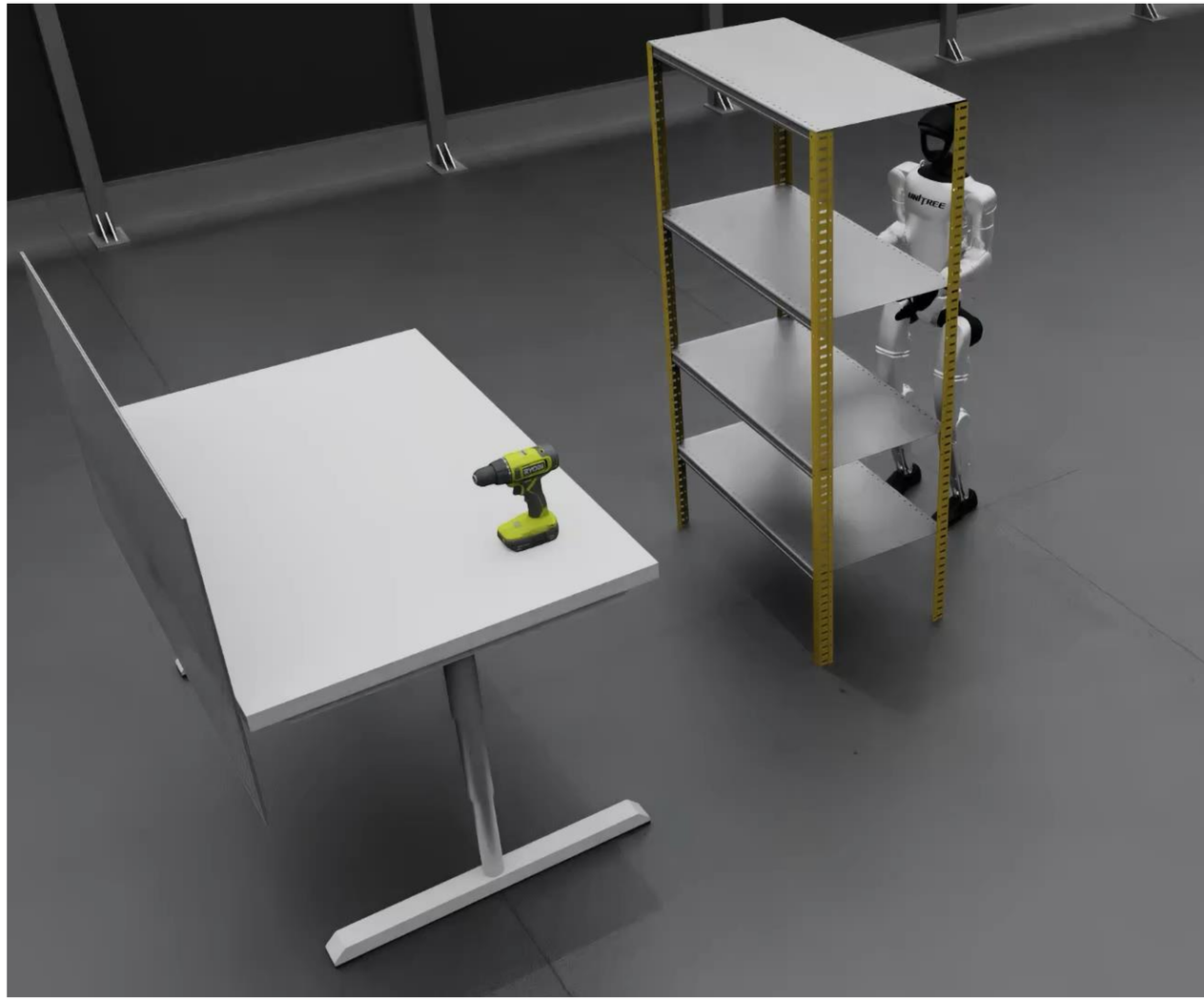
Planning Enables Demonstration Generation Across Substantial Object Pose and Environment Variance

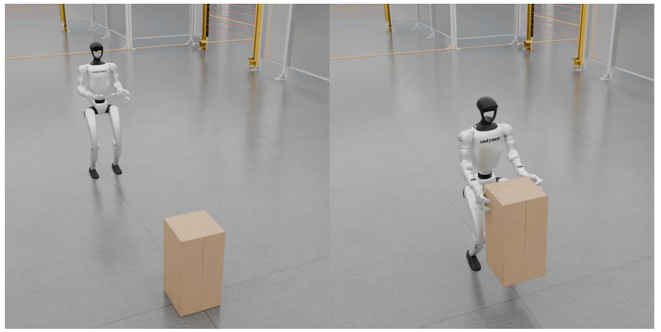


Drill Lift

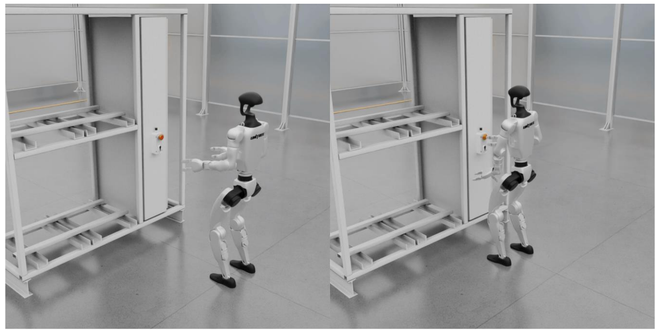


Drill Lift Obstacle

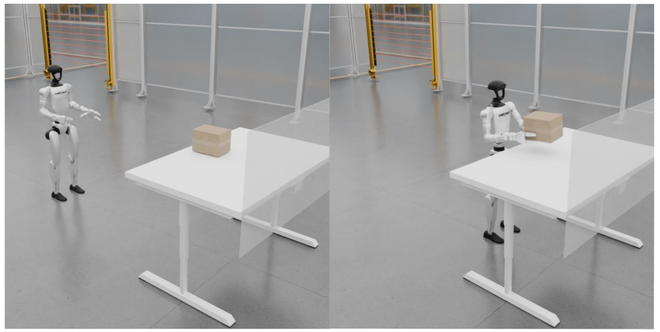
We introduce the **G1 Loco-Manipulation Benchmark**: 9 Industrial Humanoid Tasks



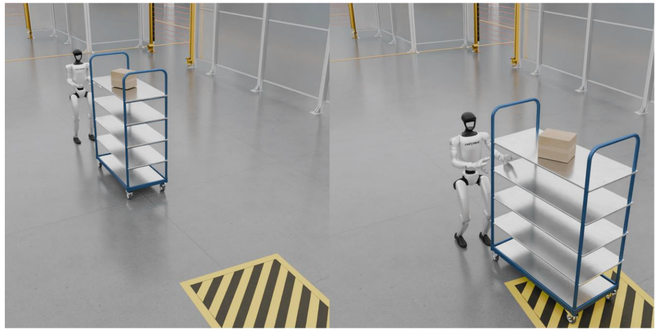
Box Lift Floor



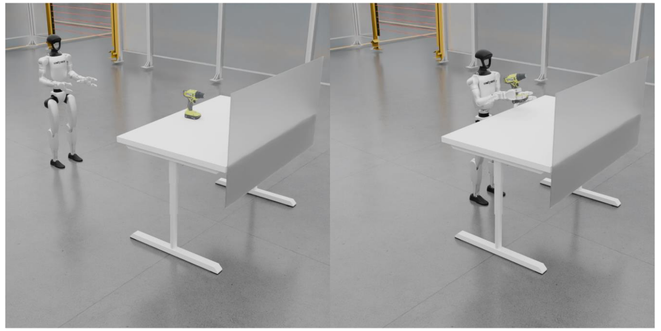
Push Button



Box Lift



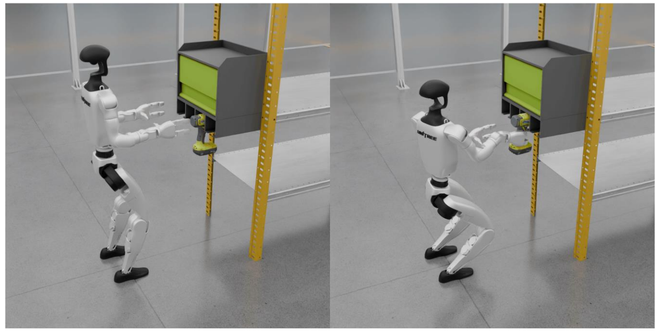
Push Shelf Forward



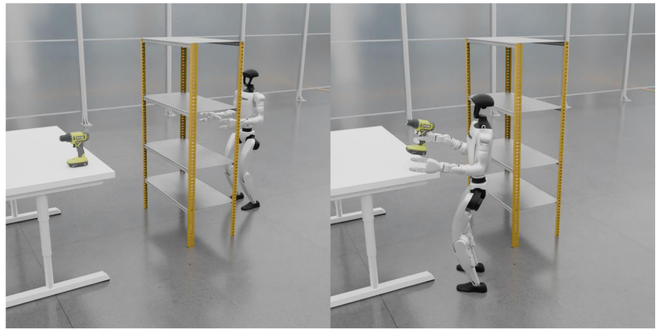
Drill Lift

Drill PnP

Box Table To Shelf



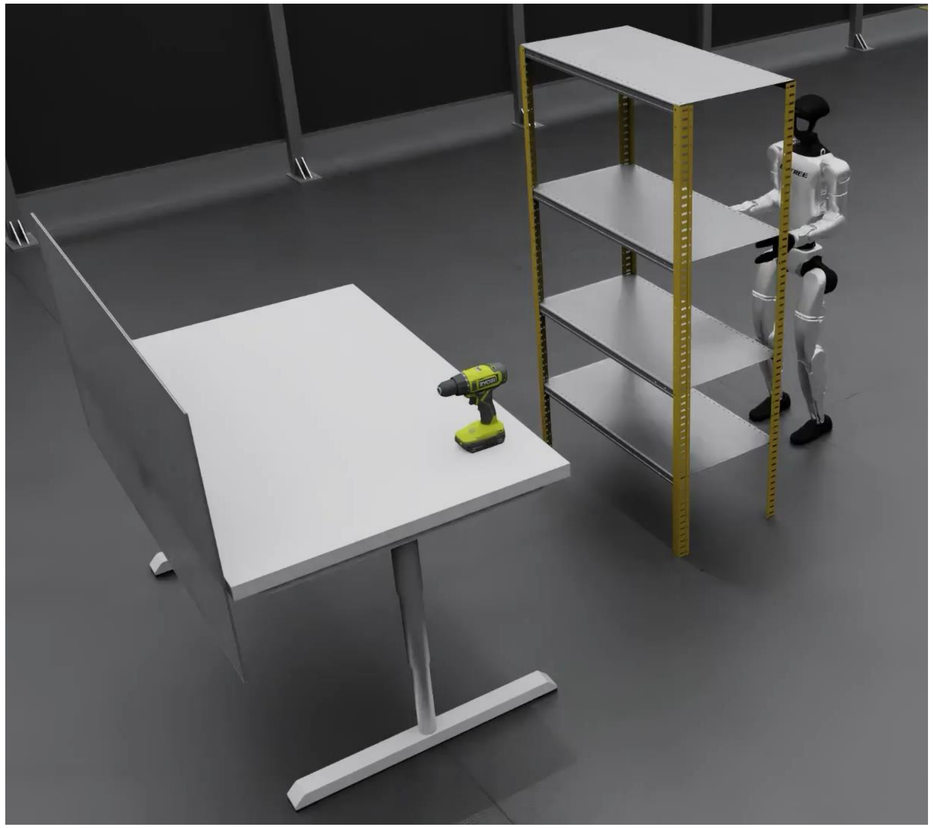
Pick Drill From Holder



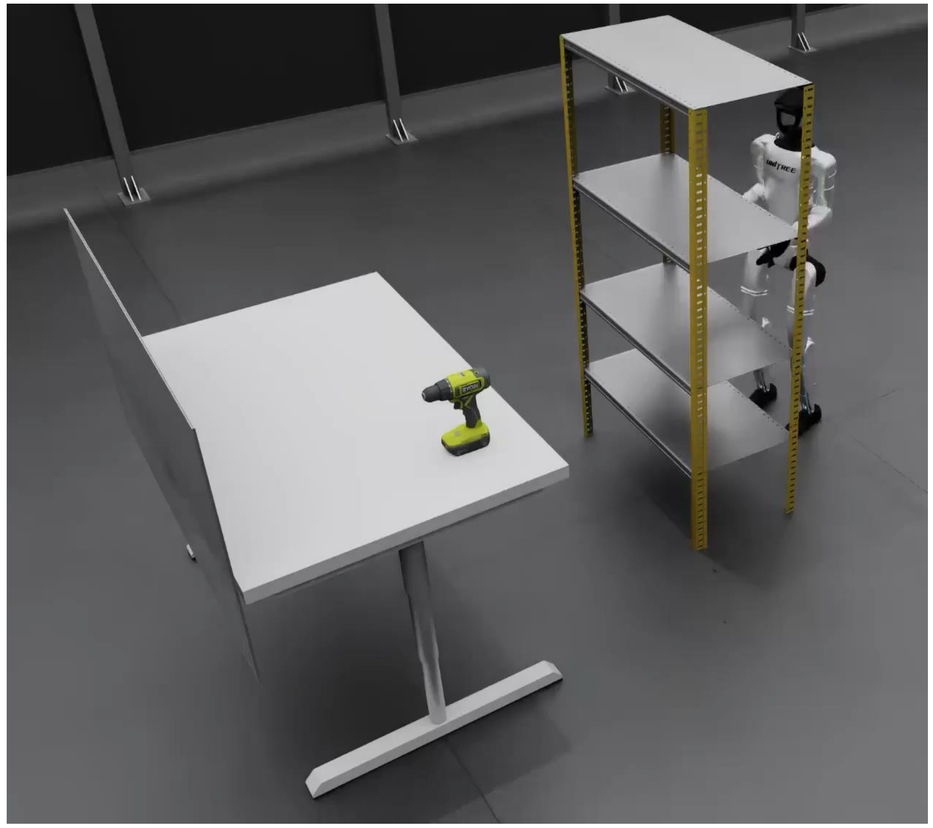
Drill Lift Obs

HumanoidMimicGen outperforms DexMimicGen+

DexMimicGen+ (DGR: 0%)

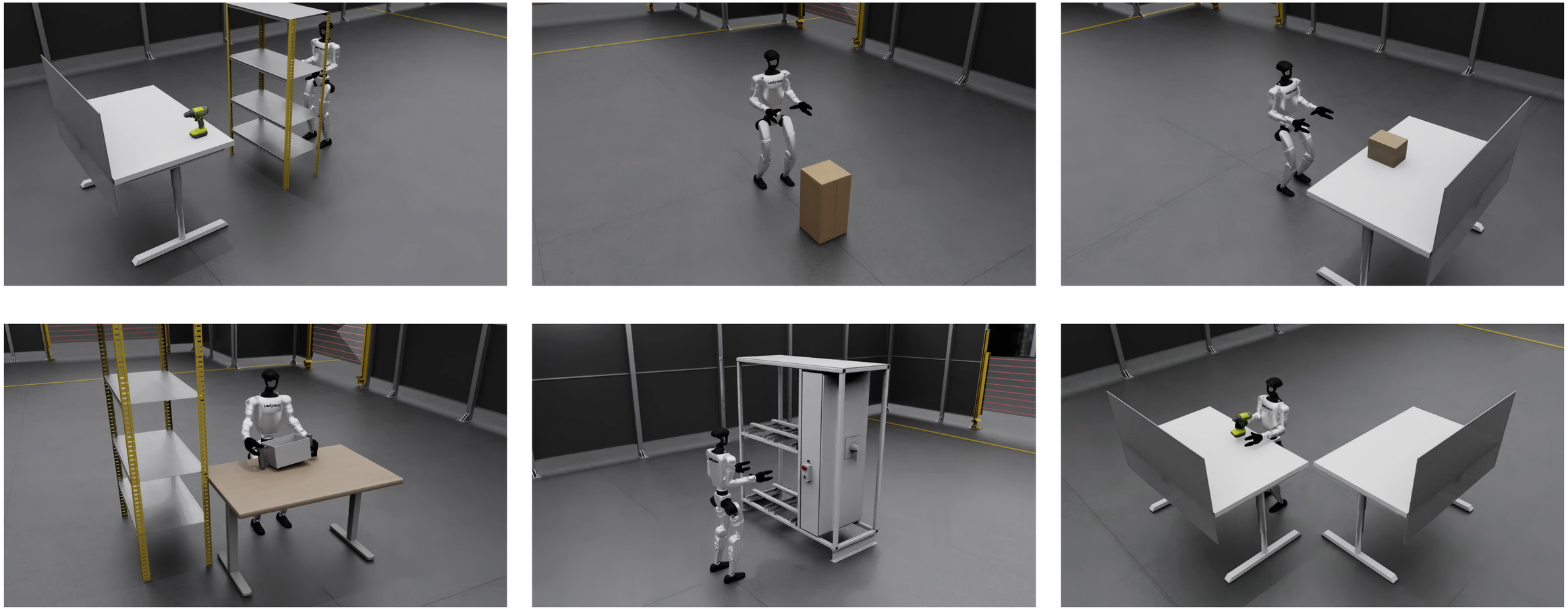


Humanoid-MG (DGR: 79%)



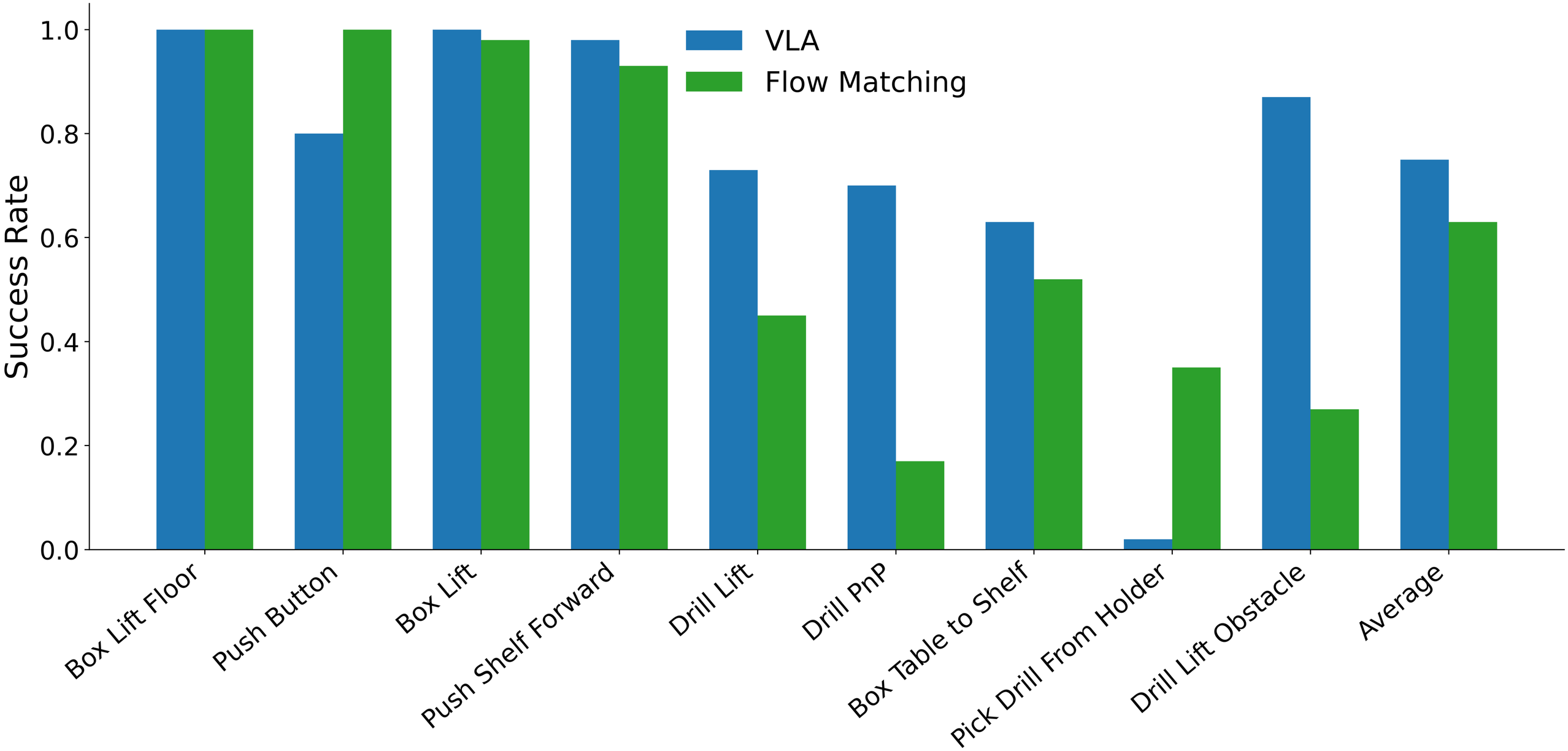
Jiang et al. DexMimicGen: Automated Data Generation for Bimanual Dexterous Manipulation via Imitation Learning. International Conference on Robotics & Automation, 2025.

HumanoidMimicGen data trains high-performing robot policies

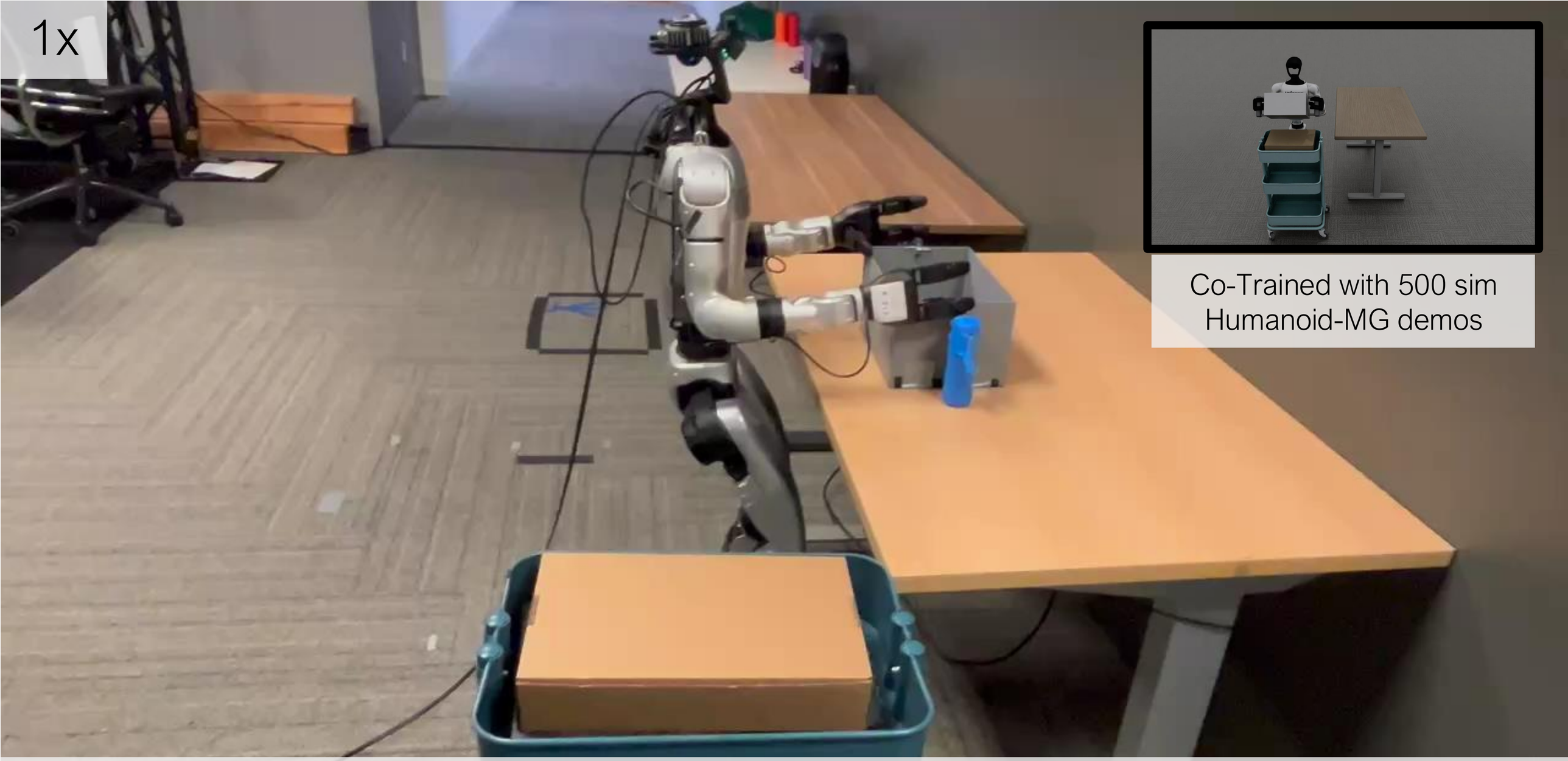


HumanoidMimicGen agents are 46% more successful than baseline method agents

HumanoidMimicGen data benefits VLAs more than simpler models



Humanoid-MG data boosts real-world performance by 20% on average



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